

Towards Scalable Reachability: Global Nonconvex Optimization of Viscous Hamilton-Jacobi PDEs

Lekan Molu[†], Haoxiang You[‡], and Ian Abraham[‡]

Abstract—Computing safe reachable sets in verification settings for high-dimensional dynamic systems, parameterized by the viscous Hamilton-Jacobi (HJ) equation, in general involves grid discretization of the phase space to achieve a numerically consistent solution with respect to the original non-viscous problem. This presents a two-fold problem: discretization implies exponential memory requirements; and, computational costs increase as problem dimensions increase. While parallel computing can reduce the impact of the latter, it does not address the grid discretization exponential memory costs. Leveraging a recent connection between HJ value functions, proximity operators, and HJ PDE envelopes, we introduce a novel moving-average Gaussian sampling scheme that computes the proximity operator for the smooth spatial gradients of the viscous HJ value in a reachability context. This circumvents nonessentially oscillatory upwinding schemes rampant in literature. It also provides a storage-free means for resolving safe sets in a computationally time-efficient manner without dealing with the exponential memory costs of grid discretization. Our hypotheses and mathematical machinery are backed up by numerical examples that communicate computational improvement schemes with clarity via statistical learning in practical safety-verification systems.

I. INTRODUCTION

This paper takes a further step towards improved computational and scalable solutions to time-dependent Hamilton-Jacobi (HJ) [1] PDEs employed in safety verification [2] of *high-dimensional* dynamical systems. We work within a viscous HJ reachability analyses context [3]–[6] and establish our results with globally optimal solutions to the reachability problems in nonlinear settings. Our concern is the first-order nonlinear scalar Hamilton-Jacobi (HJ) partial differential equations (PDE) of the form

$$\begin{aligned} v_t + H(t; x, \partial_x v) &= 0 \text{ in } \mathbb{R}^n \times (0, T] \\ v(x, 0) &= g(x) \text{ on } \mathbb{R}^n \times t = 0 \end{aligned} \quad (\text{HJ-IVP})$$

where v_t denotes the partial derivative of the vector field $v(x, t)$ with respect to t ; the Hamiltonian $H(x, t)$ is continuously defined on $\mathbb{R}^n$; $g(x)$ is bounded and uniformly continuous (BUC) in $\mathbb{R}^n$; and $\partial_x v$ is the spatial gradient of $v(x, t)$. While a unique C^2 solution $v(x, t)$ exists uniformly on $0 \leq t < T$ to (HJ-IVP) for which $\lim_{t \rightarrow T} v(x, t)$ (when H and g are smooth and $v_0(x, t)$ is compactly supported), $H(x, t)$ lacks continuous derivatives in this limit so that $\partial_x v$ vanishes at $t = T$.

However, one may find Lipschitz continuous functions $v(x, t)$ on $\mathbb{R}^n \times (0, T]$ which are smooth *almost everywhere*

(a.e.) and satisfy (HJ-IVP) in a “generalized sense”. Suppose that we fix a positive $\delta > 0$, Crandall and Lions [7] showed that if $\delta > 0$, then v^δ is the *vanishing viscosity* solution of

$$\begin{aligned} v_t^\delta + H(t; x, \nabla_x v^\delta) &= \frac{\delta}{2} \Delta v^\delta \text{ in } \mathbb{R}^n \times (0, T] \\ v^\delta(x, 0) &= g(x) \text{ on } \mathbb{R}^n \times t = 0 \end{aligned} \quad (\text{HJ-Viscous})$$

with Laplacian Δv^δ and a constant $k > 0$. Uniform convergence of $v^\delta \rightarrow v$ exists in the limit $\delta \rightarrow 0^+$, i.e.

$$\sup_{t \in (0, T)} \sup_{x \in \mathbb{R}^n} |v(x, t) - v^\delta(x, t)| \leq k\sqrt{\delta}. \quad (1)$$

Equation (HJ-Viscous) models conservation laws in a single spatial dimension [8]. For inf-sup or sup-inf optimal control problems [3], the Hamiltonian is related to the *backward* reachable set [4] of a dynamical system. Reachable systems are those where one can compute all states that can be reached from an initial condition in a *finite number of steps*.

Popular numerically consistent optimal solutions [7], [9] to (HJ-Viscous) utilize problem space grid discretization [4], [10]. However, these require exponential memory storage – making their practical applications limited for large problems. Our recent efforts [5], [6] leveraged single-instruction, multiple threads (SIMT) computing schemes on modern GPUs to accelerate computational times on a host of practical problems; we demonstrated improved scalability with up to 76 – 92% improvement in computational times on higher-dimensional problems. However, it was grid based and as problem dimensions increase, the need for more memory may make such approaches unattractive to users.

We thus tack another thread, jettisoning grid discretization methods for sampling-based proximal operators in a statistical moving average context. This resolves the global extrema of the Moreau envelopes of HJ equations and can be used to recover the reachable set in nonconvex optimization settings [11]–[14]. The Cole-Hopf transformation [1], [13] employed in PDE analysis relates the heat equation solution to that of the viscous HJ equation’s solution. Recently, Osher et. al [8] applied the Cole-Hopf transformation to v^δ and related the resulting transformed heat conservation equation to an importance sampling scheme for approximating the solution v^δ of (HJ-Viscous). We build on this, introducing a sample-efficient normal-inverse-Wishart moving-average sampling scheme for evaluating the spatial gradients of the value function in a statistical learning context so as to resolve *robust controlled backward reachable sets and tubes* [15] in the terminal Hamilton-Jacobi-Isaacs equation [4], [5], [16].

[†]Microsoft Research, 300 Lafayette Street, NYC.
[‡] Mechanical Engineering Department, Yale University.
 Email: lekanmolu@microsoft.com, {haoxiang.you, ian.abraham}@yale.edu.

The rest of this paper is structured as follows: in §II, we provide the theoretical machinery for our problem. In §III, we describe our methods. Section IV presents our numerical algorithms on sample problems and we conclude the paper in §V.

II. BACKGROUND

We first introduce the notations that are commonly throughout this article. Reachable sets within the context of two person games [16] and their accompanying “viscous” terminal HJI PDE [17] are then introduced. This is followed by the HJ-Isaacs PDEs commonly used to characterize reachable sets.

A. Notations and Terminologies

Conventions: Bold font upper-case and lower-case Roman letters are matrices and vectors, respectively; calligraphic letters are sets. Exception: the subdifferential *set* of a BUC function, $\mathbf{g}(\mathbf{x})$, is denoted ϑ . Time variables e.g. t, τ, T will always be real positive numbers. The n -dimensional Euclidean space is $\mathbb{R}^n$; the set of natural numbers is $\mathbb{N}$. The natural log of T is denoted $\log T$. The dot product is $\langle \cdot, \cdot \rangle$. The partial time derivative of $\mathbf{v}$ is $\partial \mathbf{v}_t$ while its partial spatial derivative is denoted $\partial \mathbf{v}_x$. For a function $f(x(t), y(t))$ with every argument parameterized by t , we simply write $f(t; x, y)$. We represent the real-valued Hilbert space with $\mathcal{H}$; denote $2^{\mathcal{H}}$ by its power set, i.e. the family of all subsets of $\mathcal{H}$. Associated with $\mathcal{H}$ is the maximally monotone set-valued operator, $\mathbf{Q} : \mathcal{H} \rightarrow 2^{\mathcal{H}}$. The zeroes of $\mathbf{Q}$ are denoted $\text{zer } \mathbf{Q} = \{\mathbf{x} \in \mathcal{H} \mid 0 \in \mathbf{Q}\mathbf{x}\}$. The set of global minimizers of $\mathbf{g}(\mathbf{x})$ is denoted $\text{Argmin } \mathbf{g}(\mathbf{x})$. It is written as $\text{argmin } \mathbf{g}(\mathbf{x})$ when the set is a singleton. The set of fixed point of operator $S : \mathcal{H} \rightarrow \mathcal{H}$ is $\text{Fix } S = \{\mathbf{x} \in \mathcal{H} \mid S\mathbf{x} = \mathbf{x}\}$.

The unique solution $\mathbf{x} \in \mathbb{R}^n$ of the dynamical system $\dot{\mathbf{x}}(\tau) = f(t; \mathbf{x}, \mathbf{u}, \mathbf{w})$ is influenced by a pursuing player $\mathbf{P}$ (with control $\mathbf{w} \in \mathcal{W} \subseteq \mathbb{R}^p$) and its evading pair $\mathbf{E}$ (with control $\mathbf{u} \in \mathcal{U} \subseteq \mathbb{R}^m$). System trajectories $\xi_{\mathbf{x}, \tau}^{\mathbf{u}, \mathbf{w}}$ emanate from $\dot{\mathbf{x}}(\tau)$ by minimizing a scalar-valued objective

$$\mathcal{J}(\tau; \mathbf{x}, \mathbf{u}, \mathbf{w}) = \min_{\tau \in [t, T]} \mathbf{g}(\xi_{\mathbf{x}, \tau}^{\mathbf{u}, \mathbf{w}}(\tau)) \quad (\text{Target set cost})$$

within a two-person differential game for a fixed $T \geq \tau > t$. The controllers belong in compact sets which are measurable functions

$$\bar{\mathcal{U}} \equiv \mathbf{u} : [t, T] \rightarrow \mathcal{U}, \quad \bar{\mathcal{W}} \equiv \mathbf{w} : [t, T] \rightarrow \mathcal{W}. \quad (2)$$

B. Lower value of the differential game

For a (possibly infinite) *terminal time* T , a constant k , $\mathbf{u} \in \mathcal{U}$, $\mathbf{w} \in \mathcal{W}$ and all $\hat{\mathbf{x}}, \mathbf{x} \in \mathbb{R}^n$, let the system possess a BUC payoff functional $\mathbf{g}(\mathbf{x}(T))$ that satisfies $\mathbf{g} : \mathbb{R}^n \rightarrow \mathbb{R}$. The differential game’s *lower value* [18]

$$\mathbf{v}^-(\mathbf{x}, t) = \inf_{\beta \in \mathcal{B}(t)} \sup_{\mathbf{u} \in \mathcal{U}(t)} \mathcal{J}(t; \mathbf{x}, \mathbf{u}, \mathbf{w}) \quad (3a)$$

$$\triangleq \inf_{\beta \in \mathcal{B}(t)} \sup_{\mathbf{u} \in \mathcal{U}(t)} \min_{\tau \in [t, T]} \mathbf{g}(\xi_{\mathbf{x}, \tau}^{\mathbf{u}, \mathbf{w}}(\tau)). \quad (3b)$$

is the viscosity solution to (HJ-IVP) [18, Lemma 2.1] with Hamiltonian,

$$\mathbf{H}^-(t, \mathbf{x}, \mathbf{u}, \mathbf{w}, p) = \max_{\mathbf{u} \in \mathcal{U}} \min_{\mathbf{w} \in \mathcal{W}} \langle f(t, \mathbf{x}, \mathbf{u}, \mathbf{w}), p \rangle. \quad (4)$$

Notice in (3a) that the pursuer possesses a non-anticipative strategy $\mathcal{B}(t)$ that is aware of all (maximizing) controls of the evading player up until time t . In what follows, we drop the negative superscript to refer to the value of the game in the context of reachability analysis.

C. Robustly Controlled Backward Reachable Set and Tube

Suppose that the goal of $\mathbf{E}$ is to reach a user-specified target region $\mathcal{L}_0(\tau)$ within $\tau \leq T$ time steps of playing the game; while the $\mathbf{P}$ simultaneously seeks to prevent this from happening. Since we are in a two-player best-response scenario, the invariant set $\mathcal{L}_0$ obtained at T

$$\mathcal{L}_0(T) = \{\mathbf{x} \in \mathbb{R}^n \mid \mathbf{g}(0; \mathbf{x}) \leq 0\} \quad (\text{Target-Set})$$

is “robustly controlled” for the “distance-to-target-set” cost $\mathbf{g}(0; \mathbf{x})$, with constraints $|\mathbf{g}(0; \mathbf{x})| \leq k$, $|\mathbf{g}(0; \mathbf{x}) - \mathbf{g}(t; \hat{\mathbf{x}})| \leq k|\mathbf{x} - \hat{\mathbf{x}}|$ where $k > 0$ is a constant, and all $-T \leq t \leq 0$, $\{\hat{\mathbf{x}}, \mathbf{x} \in \mathbb{R}^n\}$ ¹. The distance to $\mathcal{L}_0(\tau)$ is typically found by optimizing $\mathbf{g}(\mathbf{x}, t)$ as in (Target set cost). The HJI PDE (3a) and Hamiltonian (4) for this set admit the form

$$\begin{aligned} \mathbf{v}_t(t; \mathbf{x}) + \min\{0, \mathbf{H}(\mathbf{x}, \partial \mathbf{v}_x(t; \mathbf{x}))\} &= 0, \\ \mathbf{v}(0; \mathbf{x}) &= \mathbf{g}(0; \mathbf{x}). \end{aligned} \quad (\text{HJI-RCBRT})$$

For the safety problem setup in (3a), the corresponding *robustly controlled backward reachable tube* (RCBRT) on $(0, T]$ is the closure of the open set [15]

$$\begin{aligned} \mathcal{L}([\tau, 0], \mathcal{L}_0) &= \{\mathbf{x} \in \mathbb{R}^n \mid \exists \beta \in \bar{\mathcal{W}}(t) \forall \mathbf{u} \in \mathcal{U}(t), \exists \\ &\tau \in [-T, 0], \xi(\bar{t}) \in \mathcal{L}_0\}. \end{aligned} \quad (\text{Target-Tube})$$

Read: The set of states from which the strategies of $\mathbf{P}$ and for all controls of $\mathbf{E}$ imply that we *reach and remain in the target set* in the interval $[-T, 0]$. Following Lemma 2 of [4], the states in the reachable set admit the following properties w.r.t the value function $\mathbf{v}^-$:

$$\begin{aligned} \mathbf{x}(t) \in \mathcal{L}(\cdot) &\implies \mathbf{v}(\mathbf{x}, t) \leq 0, \\ \mathbf{v}(\mathbf{x}, t) \leq 0 &\implies \mathbf{x}(t) \in \mathcal{L}(\cdot). \end{aligned} \quad (\text{Zero Levelset})$$

In backward reach avoid tubes where the agent must avoid the unsafe region at all times, we can write (HJI-RCBRT) as the HJI variational inequality for a *robustly controlled backward reach-avoid tube* (RCBRAT) as follows,

$$\begin{aligned} \min\{\mathbf{v}_t(t; \mathbf{x}) + \mathbf{H}(t; \mathbf{x}, \partial \mathbf{v}_x), \mathbf{g}(\mathbf{x}, t) - \mathbf{v}(t; \mathbf{x})\} &\leq 0, \\ \mathbf{v}(0; \mathbf{x}) &= \mathbf{g}(0; \mathbf{x}). \end{aligned} \quad (\text{HJI-RCBRAT})$$

Challenges in optimizing the RCBRT/RCBRAT: Computing the RCBRT presents a host of challenges for optimization because (a) the dynamics $f(\cdot)$ is usually nonlinear with coupled states in most problems of interest; (b) although $\mathcal{L}([\tau, 0], \mathcal{L}_0)$ is mostly smooth, it possesses discontinuous

¹Time is reversed in BRT computational scenarios.

“shocks” at solution crossover points (see [15]); this makes global gradient-based optimization impossible for resolving v ; (c) more than one trajectory can emanate from the RCBRT’s boundary that end up at the same point on the target set – this makes sampling-based optimization difficult, but not insurmountable; and (d) the RCBRT is nonconvex.

To mitigate these challenges above, we introduce a set of tools that allow us to circumvent these challenges in an optimization context.

D. Cole-Hopf Transformation and the Heat Equation

For the BUC function $g(t; x) : \mathbb{R}^n \rightarrow \mathbb{R}$ from (HJ-IVP), consider the transformation $\varphi^\delta(x) = \exp(-v^\delta/\delta)$ originally attributed to Cole and Hopf [1, §4.4.1]. Applying the Cole-Hopf transformation to (HJ-Viscous), we can write

$$\begin{aligned} \partial \varphi_t^\delta(x, t) - \frac{\delta}{2} \varphi^\delta(x, t) &= 0 \text{ in } \mathbb{R}^n \times (0, T] \\ \varphi^\delta(x, t) &= \exp(-g/\delta) \text{ on } \mathbb{R}^n \times \{t = 0\}^2. \end{aligned} \quad (\text{Heat-Kernel})$$

The fundamental solution, $\Phi_{\delta t}$ to (Heat-Kernel) at time t is [1]

$$\Phi_{\delta t}(x) = \begin{cases} (2\pi\delta t)^{n/2} \exp\left(-\frac{|x|^2}{2\delta t}\right) & \text{in } \mathbb{R}^n \times (0, T], \\ 0 & \text{otherwise.} \end{cases}$$

The heat equation (Heat-Kernel) admits the following explicit formulae (as a convolution, an integral expression, and a Gaussian expectation; see derivation in [11, Appendix B]),

$$\varphi^\delta(t; x) = (\Phi_{\delta t} \star \exp(-g/\delta))(x), \quad (6a)$$

$$\equiv \int_{\mathbb{R}^n} (\Phi_{\delta t}(x - y) \exp(-g(y)/\delta)) dy, \quad (6b)$$

$$= \mathbb{E}_{y \sim \mathcal{N}(x, \delta t)} [\exp(-\delta^{-1} g(y))]. \quad (6c)$$

The equation (Heat-Kernel) transforms solutions to the heat equation [1], [12] to those of the viscous HJ PDE. It has enjoyed use in studying the energy landscape of neural networks [13], [19] and viscous HJ loss functions [11]. Inspecting (Heat-Kernel), we may write

$$g^\delta(x) := v^\delta(t; x) \quad (7a)$$

$$= -\delta \log(\varphi^\delta(t; x))(x) \text{ in } \mathbb{R}^n \times (0, T], \quad (7b)$$

from which we can write

$$\partial v_x^\delta(t; x) = -\delta \cdot \frac{\partial \varphi_x^\delta(t; x)}{\varphi^\delta(t; x)}. \quad (8)$$

Chaudhari et al. [13] called $g^\delta(x)$ the local entropy and postulated a stochastic differential equation (SDE) scheme that leverages $g^\delta(x)$ in SGD contexts in resolving the viscous HJ PDE. Recent publications treat HJ PDE optimization [12] in light of this local entropy function by combining Moreau envelopes [20] and proximal operator [21] algorithmics to establish zero-order optimization schemes for the viscous HJ loss function (with guaranteed convergence to global optima).

III. METHODS

We now cast the computation of the optimal “controllable”³ reachable set into a form that leverages the Moreau envelope (Moreau-Env) in a successive approximation scheme so that the reachable set can be obtained via the subdifferential operator of the problem.

A. Operators and Envelopes of HJ Equations

Monotone operator theory has had a huge impact on partial differential equations and nonlinear systems [22], [23], reducing nonlinear analyses to problems of the form

$$\text{determine } x \in \text{zer } Q = \{x \in \mathcal{H} \mid 0 \in Qx\}. \quad (9)$$

The subdifferential operator (see [21, §16]), $\partial g : \mathcal{H} \rightarrow 2^{\mathcal{H}}$, of $g(x)$ at $\bar{x}$ is the set of all ϑ that satisfies

$$\{\vartheta \in \mathcal{H} \mid (\forall x \in \mathcal{H}) \langle x - \bar{x}, \vartheta \rangle + g(\bar{x}) \leq g(x)\}. \quad (10)$$

That is, g is subdifferentiable at x if $\partial g(x) \neq \emptyset$; hence, elements of $\partial g(x)$ are the subgradients of g at x .

An optimization implication of (10) is Fermat’s rule, which states that for $g : \mathcal{H} \rightarrow (0, T)$ [24],

$$\text{argmin } g = \text{zer } \partial g. \quad (11)$$

The proximity operator of $g(x)$ at a time, $t \in (0, T]$ is the minimizers set [25]

$$\text{Prox}_{tg}(x) \triangleq \arg \min_{z \in \mathcal{H}} \left(g(z) + \frac{1}{2t} \|z - x\|^2 \right) \quad (\text{Prox-Op})$$

that define the Moreau envelope $v(\cdot, t)$, defined as [20]

$$v(x, t) = \min_{z \in \mathcal{H}} \left(g(z) + \frac{1}{2t} \|z - x\|^2 \right). \quad (\text{Moreau-Env})$$

The envelope v expands troughs of $g(x)$ and its local minimizers concur with those of g [13].

Lemma 1: Suppose that $g(x)$ is BUC, its subdifferential exists, the set $\mathcal{S}_\alpha \triangleq \{z \in \mathbb{R}^n : g(z) \leq \inf g + \alpha\}$ is compact, $\text{zer } \partial g$ exists and $x \in \mathcal{S}_\alpha$. An important result established in [11, Lemma A.3] relates the spatial gradients of the solution to (HJ-IVP) to the proximity operator of $g(x)$ at time t as

$$\text{Prox}_{tg}(x) = x - t \partial v_x(t; x) \approx x - t \cdot \partial v_x^\delta(t; x). \quad (12)$$

Our goal now is to leverage the Proximity operator in the minimization (Target set cost) while playing the zero-sum two-player differential game (HJI-RCBRAT) to optimize for the RCBRT (Target-Tube) and the zero levelset (5). In this light, we can replace $v(x, t)$ (HJ-IVP) by its viscous and smooth proxy (HJ-Viscous), evaluate the minimizer (Target set cost) using the proximal operator result (1) and solve for the inf-sup optimal control problem in (3a).

³We use this term because the agents in the differential game system have control inputs

B. The Kernel of the HJ PDE and its proximity operator

Consider the kernel $\varphi^\delta(\mathbf{x}, t)$ in (Heat-Kernel) which corresponds to the solution of (HJ-Viscous). Resolving the convolution (6) is intractable for large n and time step t . However, it is same as the probability density of a Gaussian distribution with mean $\mathbf{x}$ and variance δt (as derived in [11, Appendix B]), i.e.

$$\partial \varphi^\delta(\mathbf{x}, t) = -\frac{1}{\delta t} \cdot \mathbb{E}_{\mathbf{y} \sim \mathcal{N}(\mathbf{x}, \delta t)} [(\mathbf{x} - \mathbf{y}) \exp(\mathbf{g}(\mathbf{y})/\delta)]. \quad (13)$$

Plugging (6c) and (13) into (8), we find that

$$t \cdot \partial \mathbf{v}_\mathbf{x}^\delta(t; \mathbf{x}) = \mathbf{x} - \frac{\mathbb{E}_{\mathbf{y} \sim \mathcal{N}(\mathbf{x}, \delta t)} [\mathbf{y} \exp(-\mathbf{g}(\mathbf{y})/\delta)]}{\mathbb{E}_{\mathbf{y} \sim \mathcal{N}(\mathbf{x}, \delta t)} [\exp(-\delta^{-1} \mathbf{g}(\mathbf{y}))]} \quad (14)$$

provides the time-scaled spatial gradients of $\mathbf{v}(t; \mathbf{x})$ in (17) without the need for unstable schemes employed in upwinding methods within grid discretization numerical reachable sets' computational algorithms [5], [6], [26].

Substituting the r.h.s of (14) into (1), we find that

Proximal Operator for $\partial \mathbf{v}_\mathbf{x}^\delta(t; \mathbf{x})$

$$\text{Prox}_{t\mathbf{g}}(\mathbf{x}) = \frac{\mathbb{E}_{\mathbf{y} \sim \mathcal{N}(\mathbf{x}, \delta t)} [\mathbf{y} \cdot \exp(-\mathbf{g}(\mathbf{y})/\delta)]}{\mathbb{E}_{\mathbf{y} \sim \mathcal{N}(\mathbf{x}, \delta t)} [\exp(-\delta^{-1} \mathbf{g}(\mathbf{y}))]} \quad (15)$$

C. Hamilton-Jacobi-Proximity Operator Relation

Now consider the HJI PDEs (HJI-RCBRT) and (HJI-RCBRAT) in light of (15). We can write the domain of these PDEs in light of the proximal expression (15) as

$$\begin{aligned} \partial \mathbf{v}_t(t; \mathbf{x}) + \min\{0, \mathbf{H}(t; \mathbf{x}, \partial \mathbf{v}_\mathbf{x}(t; \mathbf{x}))\} &= 0, \\ \approx \partial \mathbf{v}_t^\delta(t; \mathbf{x}) + \min\{0, \mathbf{H}^\delta(t; \mathbf{x}, \partial \mathbf{v}_\mathbf{x}^\delta(t; \mathbf{x}))\} &= 0, \\ := \partial \mathbf{v}_t^\delta(t; \mathbf{x}) + \min \left\{ 0, \max_{\mathbf{u} \in \mathcal{U}} \min_{\mathbf{w} \in \mathcal{W}} \left\langle f(t; \mathbf{x}, \mathbf{u}, \mathbf{w}), \right. \right. \\ \left. \left. \frac{1}{t}(\mathbf{x} - \text{Prox}_{t\mathbf{g}}(\mathbf{x})) \right\rangle \right\} &= 0, \end{aligned} \quad (16)$$

for (HJI-RCBRT). Similarly, the right-hand-side of the variational inequality for (HJI-RCBRAT) can be written as

$$\begin{aligned} \min\{\partial \mathbf{v}_t(t; \mathbf{x}) + \mathbf{H}(t; \mathbf{x}, \partial \mathbf{v}_\mathbf{x}), \mathbf{g}(\mathbf{x}, t) - \mathbf{v}(t; \mathbf{x})\}, \\ \approx \min\{\partial \mathbf{v}_t^\delta(t; \mathbf{x}) + \mathbf{H}^\delta(t; \mathbf{x}, \partial \mathbf{v}_\mathbf{x}^\delta), \mathbf{g}(\mathbf{x}, t) - \mathbf{v}^\delta(t; \mathbf{x})\}, \\ := \min\{\partial \mathbf{v}_t^\delta(t; \mathbf{x}) + \max_{\mathbf{u} \in \mathcal{U}} \min_{\mathbf{w} \in \mathcal{W}} \langle f(t; \mathbf{x}, \mathbf{u}, \mathbf{w}), \partial \mathbf{v}_\mathbf{x}^\delta, \\ \mathbf{g}(t; \mathbf{x}) - \mathbf{v}^\delta(t; \mathbf{x}) \rangle\}. \end{aligned} \quad (17)$$

Thus, evaluating the spatial gradient of $\mathbf{v}^\delta$ in (17) amounts to drawing samples using the proximity expression above and using (6c) to evaluate $\mathbf{v}^\delta$. We can now write (HJI-RCBRAT)

as

$$\min \left\{ \partial \mathbf{v}_t^\delta(t; \mathbf{x}) + \max_{\mathbf{u} \in \mathcal{U}} \min_{\mathbf{w} \in \mathcal{W}} \left\langle f(t; \mathbf{x}, \mathbf{u}, \mathbf{w}), \frac{1}{t}(\mathbf{x} - \text{Prox}_{t\mathbf{g}}(\mathbf{x})) \right\rangle \right\}, \quad (18a)$$

$$\begin{aligned} \mathbf{g}(\mathbf{x}, t) + \delta \log \mathbb{E}_{\mathbf{y} \sim \mathcal{N}(\mathbf{x}, \delta t)} [\exp(-\delta^{-1} \mathbf{g}(\mathbf{y}))] \Big\} &\leq 0 \\ \mathbf{v}(0; \mathbf{x}) &= \mathbf{g}(0; \mathbf{x}). \end{aligned} \quad (18b)$$

D. Reducing Sampling Complexity

We now cast the solution of the variational HJI computational problem as a Monte Carlo algorithmics. Looking at equation (15), since we are working in a sampling-based values spatial derivative setting, we must avoid using too large samples to mitigate against time inefficiency while solving the problem.

While a weighted moving average of the gradient was mentioned in [11], no incorporation of prior data was imposed on the computational scheme. In this work, we incorporate a data-driven prior to every successive sample by first fitting Gaussian models to the numerator and denominator in (15). Preventing large samples being used to compute the proximal operator then amounts to conditioning the Gaussian model as $(\text{Prox}_{(t+1)\mathbf{g}} | \text{Prox}_{t\mathbf{g}})$ at each time step t . This is similar to linear regression with the advantage of allowing us to integrate prior information in the form of a Gaussian-inverse-Wishart distribution model.

For an equivalent sample size κ_0 and its belief, ν_0 on the prior variance, Σ_0 that is associated with the precision $\Lambda_0 = \Sigma_0^{-1}$ of a dataset D , suppose that the mean and variance of the normal-inverse-Wishart natural conjugate prior [27] are,

$$\Sigma \sim IW_{\nu_0}(\Lambda_0^{-1}), \quad \mu | \Sigma \sim \mathcal{N}(\mu_0, \Sigma/\kappa_0) \quad (19)$$

with associated empirical estimates $\hat{\Sigma}$ and $\hat{\mu}$. Then, the natural conjugate prior of the normal-inverse-Wishart is given as $p(\mu, \Sigma) \triangleq NIW(\mu_0, \kappa_0, \Lambda_0, \nu_0)$. We are concerned with its maximum-a-posteriori

$$\begin{aligned} p(\mu, \Sigma | D, \mu_0, \kappa_0, \Lambda_0, \nu_0) &= NIW(\mu, \Sigma | \mu_n, \kappa_n, \Lambda_n, \nu_n), \\ \kappa_n &= \kappa_0 + n, \quad \nu_n = \nu_0 + n \end{aligned} \quad (20)$$

where

$$\mu_n = \frac{\kappa_0 \mu_0}{\kappa_0 + n} + \frac{n}{\kappa_0 + n} \hat{\mu}, \quad (21a)$$

$$\Lambda_n = \Lambda_0 + S + \frac{\kappa_0 n (\kappa_0 + n) (\hat{\mu} - \mu_0) (\hat{\mu} - \mu_0)^\top}{\kappa_0 + n}, \quad (21b)$$

$$S = \sum_{i=1}^n (\mathbf{x} - \bar{\mathbf{x}})(\mathbf{x} - \bar{\mathbf{x}})^\top. \quad (21c)$$

We recursively incorporate new samples by leveraging the MAP parameters given in (21) to new samples drawn from (15).

IV. NUMERICAL RESULTS

In this section, we provide a representative problem and amend it to the viscous HJI PDE (HJI-RCBRT) to test and verify our hypothesis.

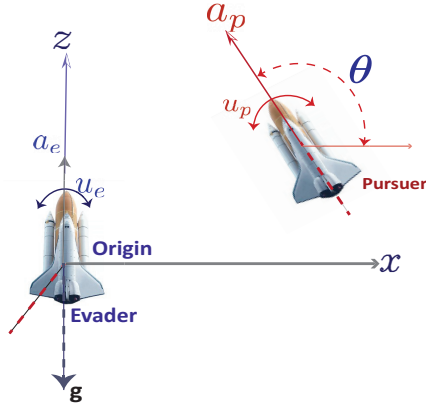


Fig. 1. Motion of two rockets on a Cartesian xz -plane with a thrust inclination in relative coordinates given by $\theta := u_p - u_e$.

A. The Rockets Launch Problem

We consider the rocket launch problem of Dreyfus [28] which was amended to a differential game between two identical rockets, P and E , on an (x, z) cross-section of a Cartesian plane in [5], [6]. We want to compute the usable part [16] or RCBRT [4] of the *approximate* terminal surface's boundary for a predefined target set over a target tube. The usable part entails the regions of the state-space for which the min-max operation over either strategy of P and E is below zero and where the boundary of the usable part (BUP) constitute where (HJI-RCBRAT) is exactly zero.

For a two-player differential game, let P and E share identical dynamics in a general sense so that we can freely choose the coordinates of P ; however, E 's origin is a distance Φ away from (x, z) at plane's origin (see Fig. 1) so that the PE vector's inclination measured counterclockwise from the x axis is θ .

Hence, we may consider the current problem as a terminal pursuit-evasion *differential* game, whereupon the game terminates when *capture* occurs. This eventual terminal represents the final *overapproximated* (non)-capture surface for the $P-E$ game (to be shortly introduced). We desire that every max and min operations during a *game of kind* are in the interior of the plane where motion exists; this way, no sudden changes from extremes are too aggravating in cost.

Let the states of P and E be (x_p, x_e) , respectively, driven by thrusts (u_p, u_e) respectively (see Figure 1). Fix the rockets' range so that what is left of the motion of either P or E 's is restricted to orientation on the (x, z) plane as illustrated in Fig. 1. where a and g are respectively the acceleration and gravitational accelerations (in feet per square second)⁴. As long as E remains within this target region or *backward reachable tube* (or BRT), P cannot cause damage or exercise an action of deleterious consequence on, say, the territory being guarded by E . Setting up E to maximize a payoff quantity with the largest possible margin or at least frustrate the efforts of P with minimal collateral damage while the pursuer minimizes this quantity

⁴We set $a = 1ft/sec^2$ and $g = 32ft/sec^2$ in our simulation.

constitutes a terminal value *optimal* differential game: there is no optimal pursuit without an optimal evasion.

The motion of P relative to E 's along the (x, z) plane includes the relative orientation, the control input, shown in Fig. 1 as $\theta = u_p - u_e$. Following the conventions in Fig. 1, the game's relative equations of motion in *reduced space* [16, §2.2] was derived as (see [5, §VI.A]) is $\mathbf{x} = (x, z, \theta)$ where $\theta \in [-\frac{\pi}{2}, \frac{\pi}{2})$ and $(x, z) \in \mathbb{R}^2$ are

$$\dot{\mathbf{x}} = \begin{cases} \dot{x} &= a_p \cos \theta + u_e x, \\ \dot{z} &= a_p \sin \theta + a_e + u_e x - g, \\ \dot{\theta} &= u_p - u_e. \end{cases} \quad (22)$$

The capture radius of the origin-centered circle Φ (we set $g = 1.5$ ft) is $\|PE\|_2$ so that $g^2 = x^2 + z^2$. All capture points are specified by the variational HJ PDE [4]:

$$\frac{\partial g}{\partial t}(\mathbf{x}, t) + \min \left[0, H\left(\mathbf{x}, \frac{\partial g(\mathbf{x}, t)}{\partial \mathbf{x}}\right) \right] \leq 0, \quad (23)$$

with Hamiltonian given by

$$H(\mathbf{x}, p) = - \max_{u_e \in [\underline{u}_e, \bar{u}_e]} \min_{u_p \in [\underline{u}_p, \bar{u}_p]} \begin{bmatrix} p_1 & p_2 & p_3 \\ a_p \cos \theta + u_e x \\ a_p \sin \theta + a_e + u_p x - g \\ u_p - u_e \end{bmatrix}. \quad (24)$$

Here, p are the co-states, and $[\underline{u}_e, \bar{u}_e]$ denotes extremals that the evader must choose as input in response to the extremal controls that the pursuer plays i.e. $[\underline{u}_p, \bar{u}_p]$. Rather than resort to analytical *geometric reasoning*, we may analyze possibilities of behavior by either agent via a principled numerical simulation. This is the essence of this work. From (24), set $\underline{u}_e = \underline{u}_p = \underline{u} \triangleq -1$ and $\bar{u}_p = \bar{u}_e = \bar{u} \triangleq +1$ so that $H(\mathbf{x}, p)$ is

$$\begin{aligned} & - \max_{u_e \in [\underline{u}_e, \bar{u}_e]} \min_{u_p \in [\underline{u}_p, \bar{u}_p]} \begin{bmatrix} p_1(a_p \cos \theta + u_e x) + \\ p_2(a_p \sin \theta + a_e + \\ u_p x - g) + p_3(u_p - u_e) \end{bmatrix} \\ & \triangleq -ap_1 \cos \theta - p_2(g - a - a \sin \theta) - \bar{u}|p_1 x + p_3| \\ & \quad + \underline{u}|p_2 x + p_3|, \end{aligned} \quad (25)$$

where the last line follows from setting $a_e = a_p \triangleq a$.

B. Numerical Integration Considerations

We take the infinitesimal physics problem above and make it algorithmic by discretizing the spatial and temporal domains i.e.

$$(x, z) \in (-100, +100], \theta \in (-\pi/2, \pi/2], t \in (0, 1], \quad (26)$$

with the resolutions $t_k - t_{k-1} = 1/1000$ along the temporal dimension and $x_k - x_{k-1} = 1/100$ along the spatial dimensions, where $k \in \mathbb{N}$.

Care is taken not to combine the state components into one full spatial domain early on in the algorithm. We first sampled from each spatial component of (26), compute the values within the expectations in (15), and then compute

the proximal for the full state space using the expectation-maximization algorithm [29]. The expectation-maximization algorithm proceeded as follows:

- We first computed the logarithm of the observations

We do not employ a mesh or grid since the indices from which we sample are uniform across x, z and θ . The innards of the expression in (15) were first computed based on samples from the state space (26).

V. CONCLUDE

REFERENCES

- [1] L. C. Evans, *Partial Differential Equations*. American Mathematical Society, 2022, vol. 19. 1, 3
- [2] K. L. Hobbs, M. L. Mote, M. C. Abate, S. D. Coogan, and E. M. Feron, “Runtime assurance for safety-critical systems,” *IEEE Control Systems Magazine*, vol. 43, pp. 28–65, 2023. 1
- [3] J. Lygeros, “On reachability and minimum cost optimal control,” *Automatica*, vol. 40, no. 6, pp. 917–927, 2004. 1
- [4] I. M. Mitchell, A. M. Bayen, and C. J. Tomlin, “A Time-Dependent Hamilton-Jacobi Formulation of Reachable Sets for Continuous Dynamic Games,” *IEEE Transactions on Automatic Control*, vol. 50, no. 7, pp. 947–957, 2005. 1, 2, 5
- [5] “The python levelset toolbox (levelsetpy),” in *IEEE 63rd Conference on Decision and Control (CDC)*, 2024, pp. 8938–8945. 1, 4, 5
- [6] L. Molu, “The python levelset software package,” *The ACM Transactions on Mathematical Software*, 2025. 1, 4, 5
- [7] M. G. Crandall and P.-L. Lions, “Two Approximations of Solutions of Hamilton-Jacobi Equations,” *Mathematics of Computation*, vol. 43, no. 167, pp. 1–19, 1984. 1
- [8] S. Osher and C.-W. Shu, “High-Order Essentially Nonoscillatory Schemes for Hamilton-Jacobi Equations,” *SIAM Journal of Numerical Analysis*, vol. 28, no. 4, pp. 907–922, 1991. 1
- [9] M. Crandall and A. Majda, “The method of fractional steps for conservation laws,” *Numerische Mathematik*, vol. 34, no. 3, pp. 285–314, 1980. 1
- [10] S. Osher and R. Fedkiw, “Level Set Methods and Dynamic Implicit Surfaces,” *Applied Mechanics Reviews*, vol. 57, no. 3, pp. B15–B15, 2004. 1
- [11] H. Heaton, S. Wu Fung, and S. Osher, “Global solutions to nonconvex problems by evolution of hamilton-jacobi pdes,” *Communications on Applied Mathematics and Computation*, vol. 6, no. 2, pp. 790–810, 2024. 1, 3, 4
- [12] M. G. Crandall and P.-L. Lions, “A hamilton-jacobi-based proximal operator,” *Mathematics of computation*, vol. 43, no. 167, pp. 1–19, 2023. 1, 3
- [13] P. Chaudhari, A. Oberman, S. Osher, S. Soatto, and G. Carlier, “Deep relaxation: partial differential equations for optimizing deep neural networks,” *Research in the Mathematical Sciences*, vol. 5, pp. 1–30, 2018. 1, 3
- [14] P. Chaudhari, A. Choromanska, S. Soatto, Y. LeCun, C. Baldassi, C. Borgs, J. Chayes, L. Sagun, and R. Zecchina, “Entropy-sgd: Biasing gradient descent into wide valleys,” *Journal of Statistical Mechanics: Theory and Experiment*, vol. 2019, no. 12, p. 124018, 2019. 1
- [15] I. Mitchell, “A Robust Controlled Backward Reach Tube with (Almost) Analytic Solution for Two Dubins Cars,” *EPiC Series in Computing*, vol. 74, pp. 242–258, 2020. 1, 2, 3
- [16] R. Isaacs, *Differential Games: A Mathematical Theory with Applications to Warfare and Pursuit, Control and Optimization*. Kreiger, Huntigton, NY, 1999. 1, 2, 5
- [17] L. Evans and P. E. Souganidis, “Differential Games And Representation Formulas For Solutions Of Hamilton-Jacobi-Isaacs Equations,” *Indiana Univ. Math. J.*, vol. 33, no. 5, pp. 773–797, 1984. 2
- [18] —, “Differential games and representation formulas for solutions of Hamilton-Jacobi-Isaacs equations,” *Indiana Univ. Math. J.*, vol. 33, no. 5, pp. 773–797, 1984. 2
- [19] C. Baldassi, A. Ingrosso, C. Lucibello, L. Saglietti, and R. Zecchina, “Local entropy as a measure for sampling solutions in constraint satisfaction problems,” *Journal of Statistical Mechanics: Theory and Experiment*, vol. 2016, no. 2, p. 023301, 2016. 3
- [20] J. Moreau, “Décomposition Orthogonale d’un Espace Hilbertien Selon deux Cônes Mutuellement Polaires,” *Comptes rendus hebdomadaires des séances de l’Académie des sciences*, vol. 255, p. 238–240, 1962. 3
- [21] H. H. Bauschke, P. L. Combettes, H. H. Bauschke, and P. L. Combettes, *Convex analysis and monotone operator theory in Hilbert spaces*. Springer, 2017. 3
- [22] E. H. Zarantonello, “Projections on Convex Sets in Hilbert Space and Spectral Theory,” in *Contributions to Nonlinear Functional Analysis*. Elsevier, 1971, pp. 237–424. 3
- [23] R. I. Boţ and C. Hendrich, “Convergence Analysis for a Primal-dual Monotone + Skew Splitting Algorithm with Applications to Total Variation Minimization,” *Journal of mathematical imaging and vision*, vol. 49, no. 3, pp. 551–568, 2014. 3
- [24] J.-J. Moreau, “Proximity and Duality in a Hilbertian space,” *Bulletin of the Mathematical Society of France*, vol. 93, pp. 273–299, 1965. 3
- [25] J. J. Moreau, “Dual convex functions and proximal points in a hilbert space,” *Weekly reports of the sessions of the Academy of Sciences*, vol. 255, pp. 2897–2899, 1962. 3
- [26] I. Mitchell, “A toolbox of level set methods, version 1.0,” *The University of British Columbia, UBC CS TR-2004-09*, pp. 1–94, July 2004. 4
- [27] K. P. Murphy, “Conjugate bayesian analysis of the gaussian distribution,” *Notes, University of British Columbia, Department of Computer Science*, p. 16, 2007. 4
- [28] S. E. Dreyfus, “Control Problems With Linear Dynamics, Quadratic Criterion, and Linear Terminal Constraints,” Rand Corp, Santa Monica Calif, Tech. Rep., 1966. 5
- [29] A. P. Dempster, N. M. Laird, and D. B. Rubin, “Maximum likelihood from incomplete data via the em algorithm,” *Journal of the royal statistical society: series B (methodological)*, vol. 39, no. 1, pp. 1–22, 1977. 6